Netflix -themed MySQL assignment

Database Name: NetflixDB

Tables and Schema:

Users:

- user_id INT PRIMARY KEY AUTO_INCREMENT,
- name VARCHAR(100) NOT NULL,
- email VARCHAR(150) UNIQUE NOT NULL,
- registration_date DATE NOT NULL,
- plan ENUM('Basic', 'Standard', 'Premium') DEFAULT 'Basic'

Movies:

- movie_id INT PRIMARY KEY AUTO_INCREMENT,
- title VARCHAR(200) NOT NULL,
- genre VARCHAR(100) NOT NULL,
- release_year YEAR NOT NULL,
- rating DECIMAL(3, 1) NOT NULL

WatchHistory:

- watch_id INT PRIMARY KEY AUTO_INCREMENT,
- user_id INT,
- movie_id INT,
- watched_date DATE NOT NULL,
- completion_percentage INT CHECK (completion_percentage >= 0 AND completion_percentage <= 100),
- FOREIGN KEY (user_id) REFERENCES Users(user_id),
- FOREIGN KEY (movie_id) REFERENCES Movies(movie_id)

Reviews:

- review_id INT PRIMARY KEY AUTO_INCREMENT,
- movie_id INT,
- user_id INT,
- review_text TEXT,
- rating DECIMAL(2, 1) CHECK (rating >= 0 AND rating <= 5),
- review_date DATE NOT NULL,
- FOREIGN KEY (movie_id) REFERENCES Movies(movie_id),
- FOREIGN KEY (user_id) REFERENCES Users(user_id)

Data Insertion:

Users:

INSERT INTO Users (name, email, registration_date, plan)

VALUES

('John Doe', 'john.doe@example.com', '2024-01-10', 'Premium'),

('Jane Smith', 'jane.smith@example.com', '2024-01-15', 'Standard'),

('Alice Johnson', 'alice.johnson@example.com', '2024-02-01', 'Basic'),

('Bob Brown', 'bob.brown@example.com', '2024-02-20', 'Premium');

Movies:

INSERT INTO Movies (title, genre, release_year, rating)

VALUES

('Stranger Things', 'Drama', 2016, 8.7),

('Breaking Bad', 'Crime', 2008, 9.5),

('The Crown', 'History', 2016, 8.6),

('The Witcher', 'Fantasy', 2019, 8.2),

('Black Mirror', 'Sci-Fi', 2011, 8.8);

WatchHistory:

INSERT INTO WatchHistory (user_id, movie_id, watched_date, completion_percentage)

VALUES

(1, 1, '2024-02-05', 100),

(2, 2, '2024-02-06', 80),

(3, 3, '2024-02-10', 50),

(4, 4, '2024-02-15', 100),

(1, 5, '2024-02-18', 90);

Reviews:

INSERT INTO Reviews (movie_id, user_id, review_text, rating, review_date)

VALUES

(1, 1, 'Amazing storyline and great characters!', 4.5, '2024-02-07'),

(2, 2, 'Intense and thrilling!', 5.0, '2024-02-08'),

(3, 3, 'Good show, but slow at times.', 3.5, '2024-02-12'),

(4, 4, 'Fantastic visuals and acting.', 4.8, '2024-02-16');

Assignment Questions:

1. List all users subscribed to the Premium plan:
2. Retrieve all movies in the Drama genre with a rating higher than 8.5:

3. Find the average rating of all movies released after 2015:
4. List the names of users who have watched the movie Stranger Things along with their completion percentage:
5. Find the name of the user(s) who rated a movie the highest among all reviews:
6. Calculate the number of movies watched by each user and sort by the highest count:
7. List all movies watched by John Doe, including their genre, rating, and his completion percentage:
8. Update the movie's rating for Stranger Things:
9. Remove all reviews for movies with a rating below 4.0:
10. Fetch all users who have reviewed a movie but have not watched it completely (completion percentage < 100):
11. List all movies watched by John Doe along with their genre and his completion percentage:
12. Retrieve all users who have reviewed the movie Stranger Things, including their review text and rating:
13. Fetch the watch history of all users, including their name, email, movie title, genre, watched date, and completion percentage:
14. List all movies along with the total number of reviews and average rating for each movie, including only movies with at least two reviews: